

Course Title	Microprocessor Interfacing & Programming	Course Code	EE3002
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering Practice (WK6)	Credit Hrs.	3+1
Knowledge Area	Computer Engineering (KA01)	Grading Scheme	Relative
HEC Knowledge Area	Depth Electives	Applicable From	Spring 2023
Pre-requisite(s)	EE/EL1005 Digital Logic Design		

Course Objective	The course introduces the architecture, programming, and interfacing of microprocessor/microcontroller-based systems. A portion of this course comprises of assembly/C language programming to understand the relation of software and hardware in microprocessor/microcontroller-based systems. At the end of the course the student should be able to utilize a set of concepts common to design and implement stand-alone microcontroller-based systems which includes design, analysis of design, development of software and hardware required, and testing of the system using different available tools.
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No.	Assigned Program Learning Outcome (PLO)
3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

I = Introduction, R = Reinforcement E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Describe the internal architecture, programming of microprocessor and microcontroller.	A1, Q1, M1	C2	3
2	Analyze the low-level programming and interfacing challenges associated with the microprocessor/microcontroller	A2, Q2, M2	C4	3
3	Apply software implementation and make use of necessary steps required to interface a microcontroller with various hardware components.	Q3, M2, F	C3	3
4	Design practical system using Intel's micro-processor and micro-controller.	A3, F	C5	3

Text Books	Title	The Intel Microprocessors (8th Edition)
	Author	Berry. B. Brey
	Publisher	Pearson; 8th edition (June 28, 2008) ISBN-10: 0135026458
	Title	The PIC microcontrollers
	Author	Nebojsa Matic
	Publisher	mikroElektronika, 2003 (Free Available Online)
Reference Books	Title	Microprocessors and Interfacing
	Author	Douglas V. Hall
	Publisher	McGraw-Hill, 2006, 2nd Edition, ISBN 0-07-060167-4

Week	Course Contents/Topics	Chapter	CLO
01	Introduction to computer architecture, microprocessor 8086 and Addressing Modes	1, 2 (TB-1)	1
02	Software model, Instruction Set (Data Transfer instructions 8086)	3,4 (TB-1)	1,2
03	8086 Instruction set (Arithmetic and Logic instructions)	3,4 (TB-1)	1,2
04	x86 Program flow Instructions, I/O and Hardware Specifications	5,6,8 (TB-1)	1
05-06	General Microcontroller Architecture and PIC Microcontroller Architecture	1, 2 (TB-2)	1
07	Instruction Set (Data transfer, Arithmetic) PIC microcontroller	3 (TB-2)	2
08	Instruction Set (Logic, Shift Rotate instructions) PIC microcontroller	3 (TB-2)	2
09	Instruction Set (Program Flow instructions) PIC microcontroller	3 (TB-2)	2,3
10	I/O configuration and Interfacing (PORTS of PIC microcontroller)	4,6 (TB-2)	2,3
11	Timers and Control units of Controller (PIC microcontroller)	4,6 (TB-2)	3,4
12 – 13	Interrupts (External, internal, and other interrupts of PIC microcontroller)	4,6 (TB-2)	3,4
14 – 15	Reading and writing to EEPROM (PIC microcontroller)	4,6 (TB-2)	2,3
16	Writing mix Assembly and C code and Project Building using PIC microcontroller	4,6 (TB-2)	4

***TB-1 stands for Text book (The Intel Microprocessors)**

Assessment Tools	Weightage
Quizzes, Assignments*	20%
Midterm (I+II)	15+15 = 30%
Final Exam	50%

*Assignments/Quiz will also comprise of a final hardware based project